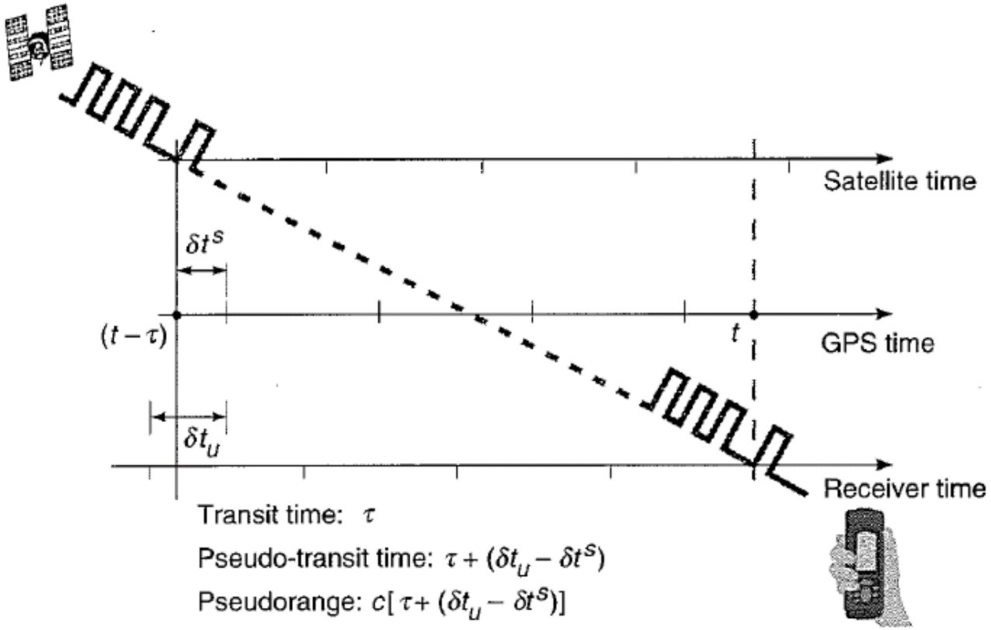
ASEN 5090 Lecture 3
GPS Signal Structure



Week	Date	Topic	Reading	Assignments - Due 11pm*
1	8/24/26	L1. Introduction & GPS Basics	PNT Ch 1, ME 2.1 & 2.2	
	8/26/26	L2. Navigation Measurements & Methods	ME Ch 1	
	8/28/26			HW1 Assigned
2	8/31/26	L3. GPS Signal Structure	ME Ch 2, PNT Ch 2	
	9/2/26	L4. Coordinate Frames	ME Ch 4.1, 4.A	
	9/4/26			HW1 Due, HW2 Assigned
3	9/7/26	Labor Day (no class)		
	9/9/26	L5. Orbits & Visibility	ME Ch 4.3	
	9/11/26			HW3 Assigned

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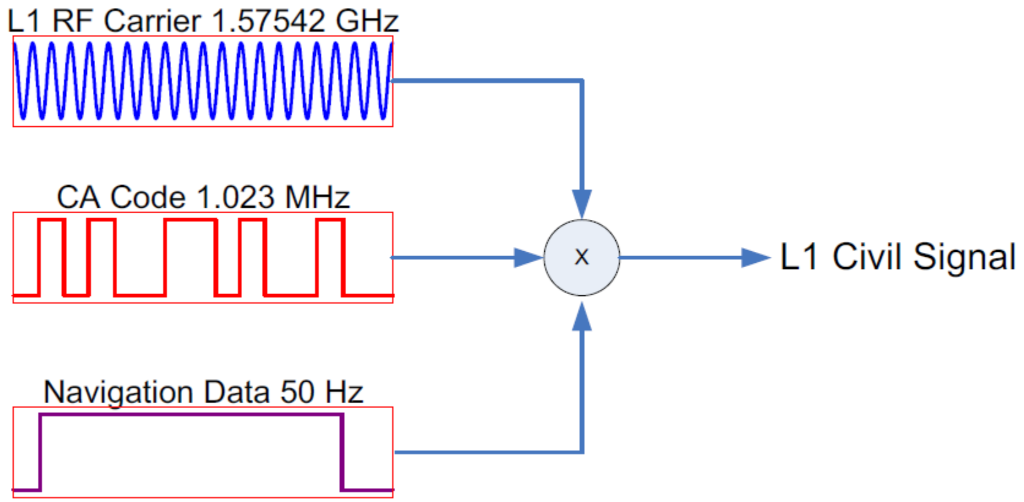
Outline

- GPS Signal Components – focus on legacy signals
- Carriers
- Codes – Purpose, characteristics, generation, modulation onto carriers
- Navigation data – Purpose, content, organization, combining with codes

- **Reading Assignment**
  - ME Ch 2, problems 2-1 and 2-2 and PNT Ch 2
- **Questions to think about**
  - What features of GPS are enabled by the codes?
  - Why 2 frequencies?
  - How does a receiver find a specific GPS satellite?

GPS Signal Components



Base frequency is f_0 = 10.23 MHz = 10,230,000 Hz (cycles/sec)

Carriers – Sinusoids

- L1 = 154 f_0 = 1575.42 MHz
- L2 = 120 f_0 = 1227.6 MHz
- L5 = 115 f_0 = 1176.45 MHz

Codes – Known sequence of 0's and 1's, transmitted at known times

Legacy Codes:

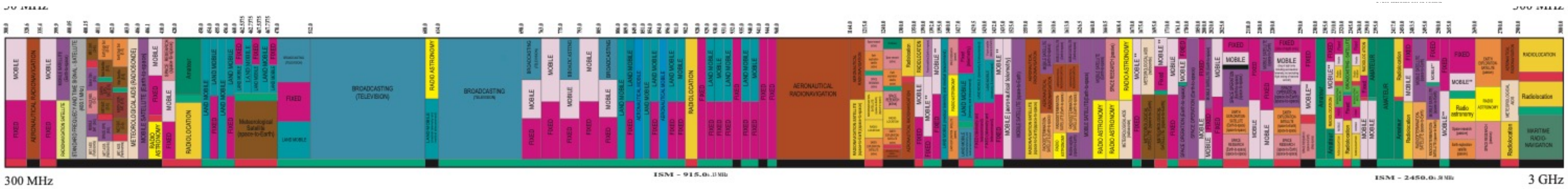
- C/A code: **chip rate** = f_0 /10 = 1.023 Mcps (MHz), **length** = 1 ms
- P code: chip rate = f_0 = 10.23 Mcps (MHz), 37 weeks long, each satellite gets a 1 week segment

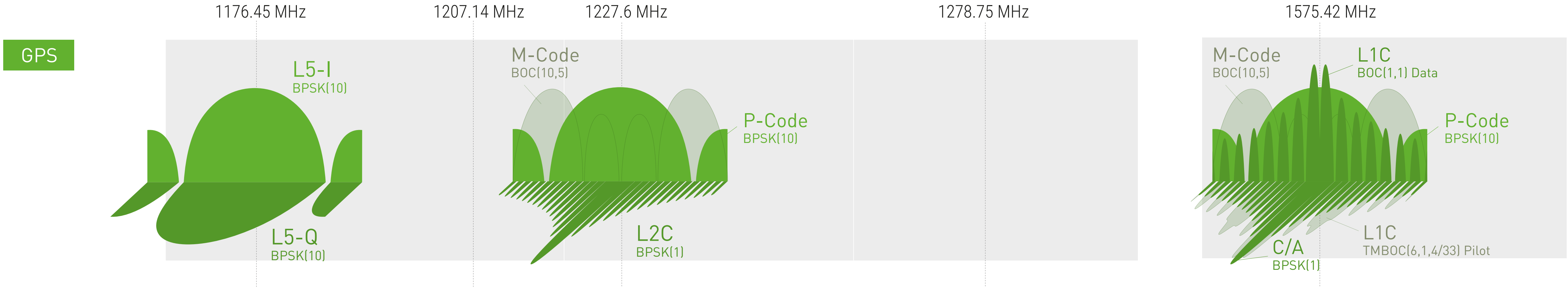
Legacy Navigation Data

- Data rate = 50 bps, 20ms per bit, 6 s per subframe, 30 s for individual satellite data
  12.5 min for all

GPS Spectrum

UNITED
STATES
FREQUENCY
ALLOCATIONS
THE RADIO SPECTRUM





From https://www.orolia.com/wp-content/uploads/2021/07/GNSS_Spectrum_11x17_11-14-2019-1.pdf

CODES: Three Functions of PRN Codes

1. **Satellite ID**
   - Each satellite transmits a different PRN code sequence

2. **CDMA – Code Division Multiple Access**
   - Interference suppression among satellite signals
   - PRN code spreads the spectrum of each signal into a wider bandwidth
   - Receiver recovers a specific satellite's signal by de-spreading process

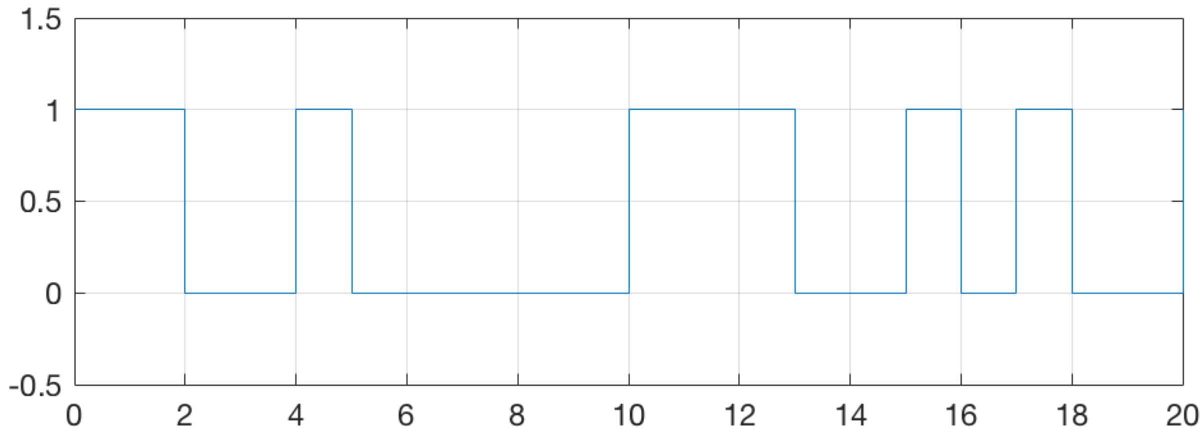
3. **Receiver-Satellite range measurement**
   - PRN code phase is a measure of signal transmission time

What is a PRN code?

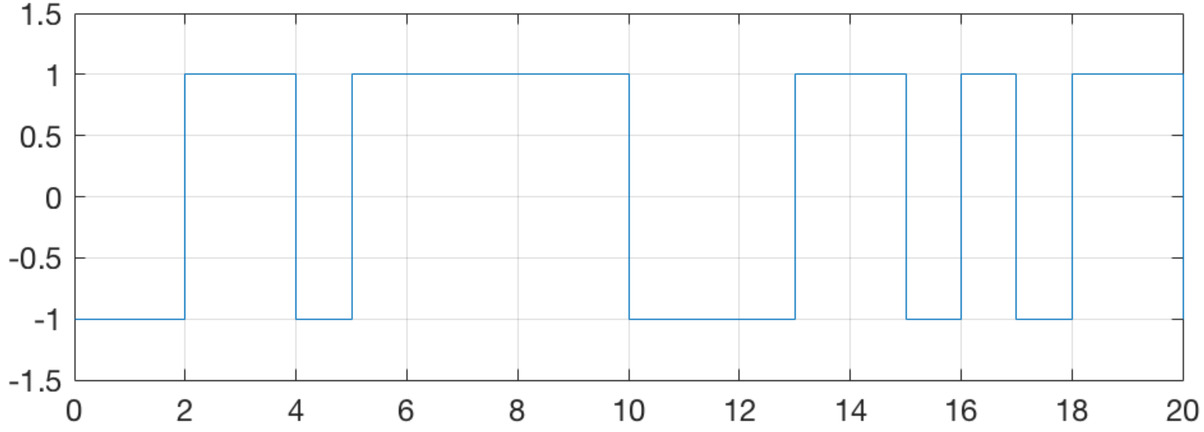
Noise-like digital sequence:

- Deterministic
- Important auto & cross correlation properties
- Has a large bandwidth or spread spectrum

Code as 0's and 1's



Code as +1/-1



CORRELATION FUNCTION Rxy(τ)

- Correspondence between a code and a phase-shifted replica of another code
- Cross correlation – correspondence between different codes: Rxy(τ)
- Autocorrelation - correspondence of code w/phase shifted version of itself Rx(τ)

$$R^{j,k}\left(n\right)=\sum_{i=1}^{1023}x^{j}\left(i\right)x^{k}\left(i+n\right),\ \ \text{for good cross-correlation}\approx 0, j\neq k$$

$$R\left(n\right)=\sum_{i=1}^{1023}x^{j}\left(i\right)x^{j}\left(i+n\right),\ \ \text{for good auto-correlation}\approx 0, n\neq 0$$

- Simple idea: Compare shifted code replica with the original reference.
  If just a binary code, number of bits that agree minus the number of bits that disagree is the correlation.

CORRELATION CALCULATION - EXAMPLES

Cross correlation between two different codes for τ =0 (no shift):

100111101110100010011011111111101

000010001010011100001110010010001

Every time the numbers agree, add 1.

Every time the numbers disagree, subtract 1.

100111101110100010011011111111111

000010001010011100001110010010001

14 agree

19 disagree

Total score: -5

Perfect agreement would be 33

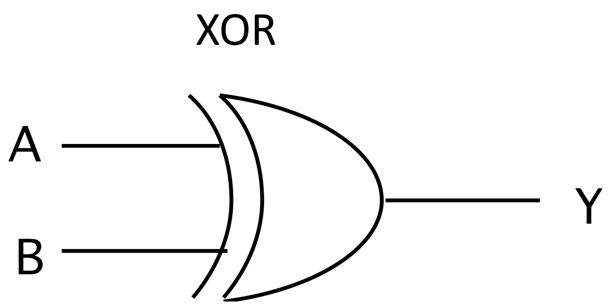
Normalized correlation: -5/33

Calculating Autocorrelation R(τ)

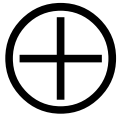
SHIFT	CODE	AGREE	DISAGREE	AUTOCORRELATION
REF	1110010	-	-	-
0	1110010	7	0	7
1	0111001			
2				
3				
4				
5				
6				
7				

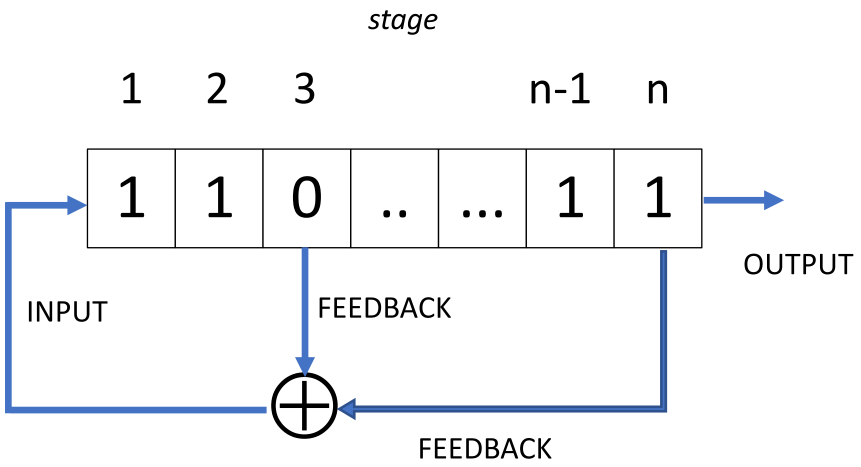
How are PRN codes generated?

- Tapped Feedback Shift Register



A	B	Y
0	0	0
1	0	1
0	1	1
1	1	0





n stage shift register tapped at [3,n]

GENERATING A CODE

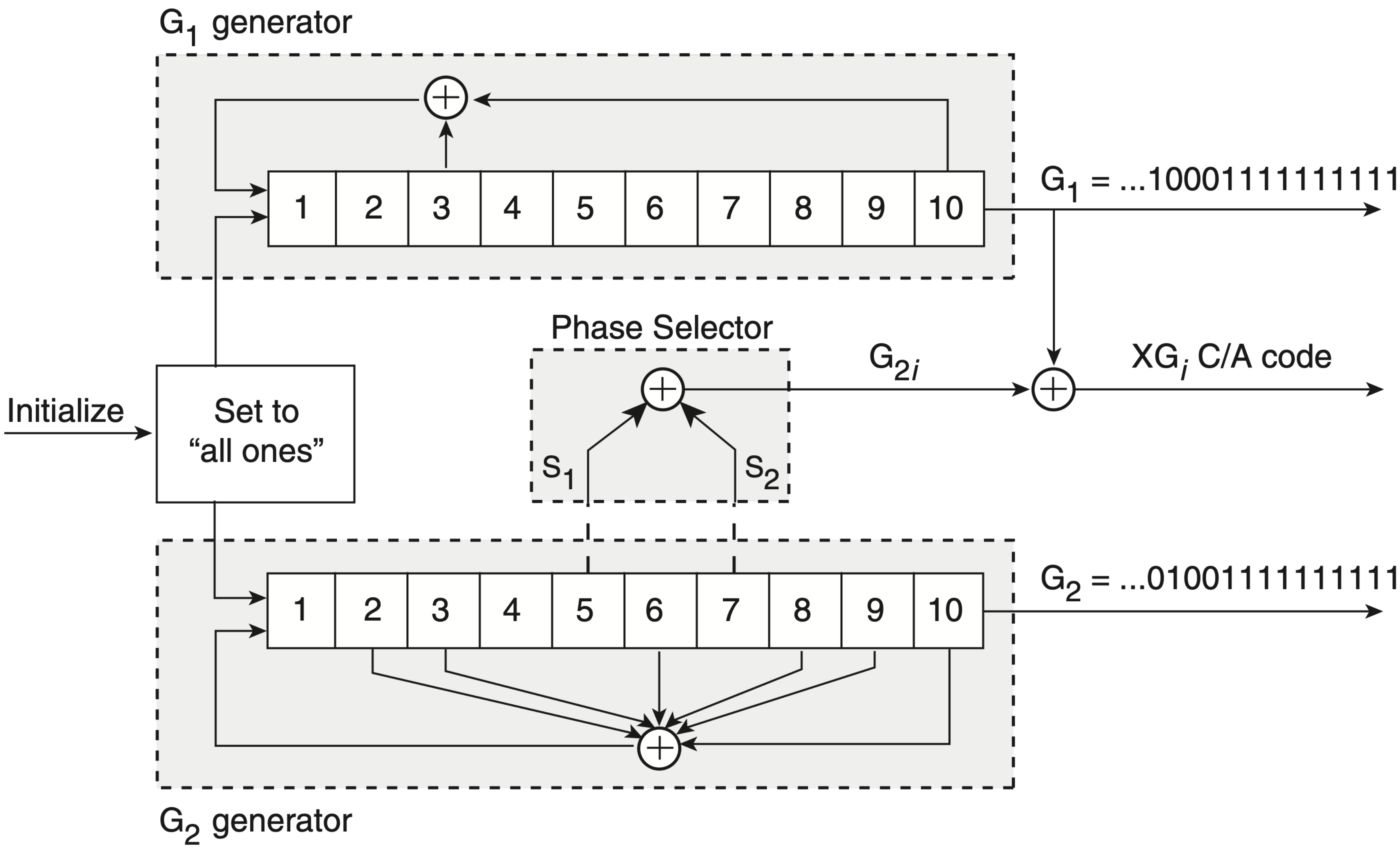
- Example: Generate the PRN code for a 3 stage shift register tapped at [3,2] starting from an all ones state.

Shift	Stage			Code
T	1	2	3	
0	1	1	1	
1				
2				
3				
4				
5				
6				
7				

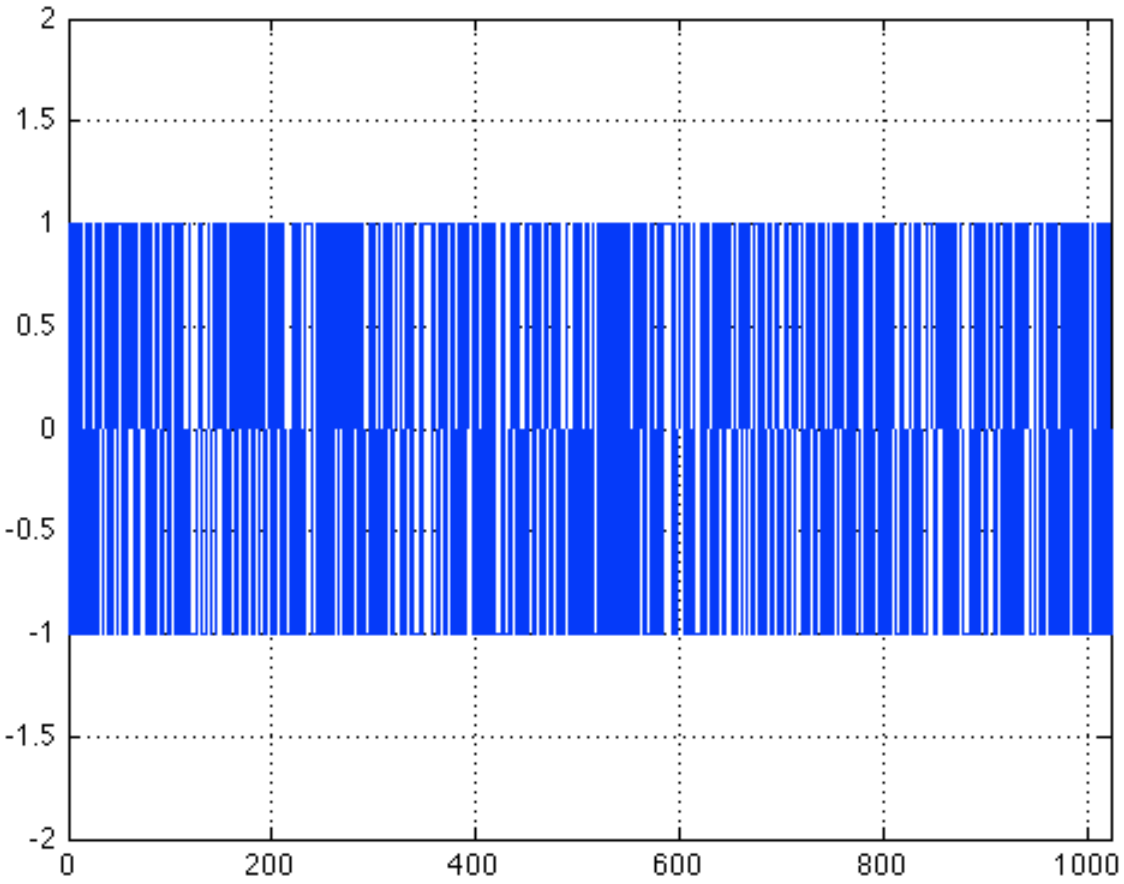
MAXIMAL LENGTH CODES

- Maximal length codes are the longest codes that can be generated by a given number of shift register stages. For an *n*-stage shift register, the longest code is $2^n - 1$ chips (i.e. 1's or 0's).
- Maximal length codes have many interesting properties, especially their autocorrelation function.
- For a maximal length code, the autocorrelation function for a phase shift other than 0 or $(2^n - 1)$ is -1.
- For any code, the autocorrelation for a 0 shift is equal to the number of chips in the code.

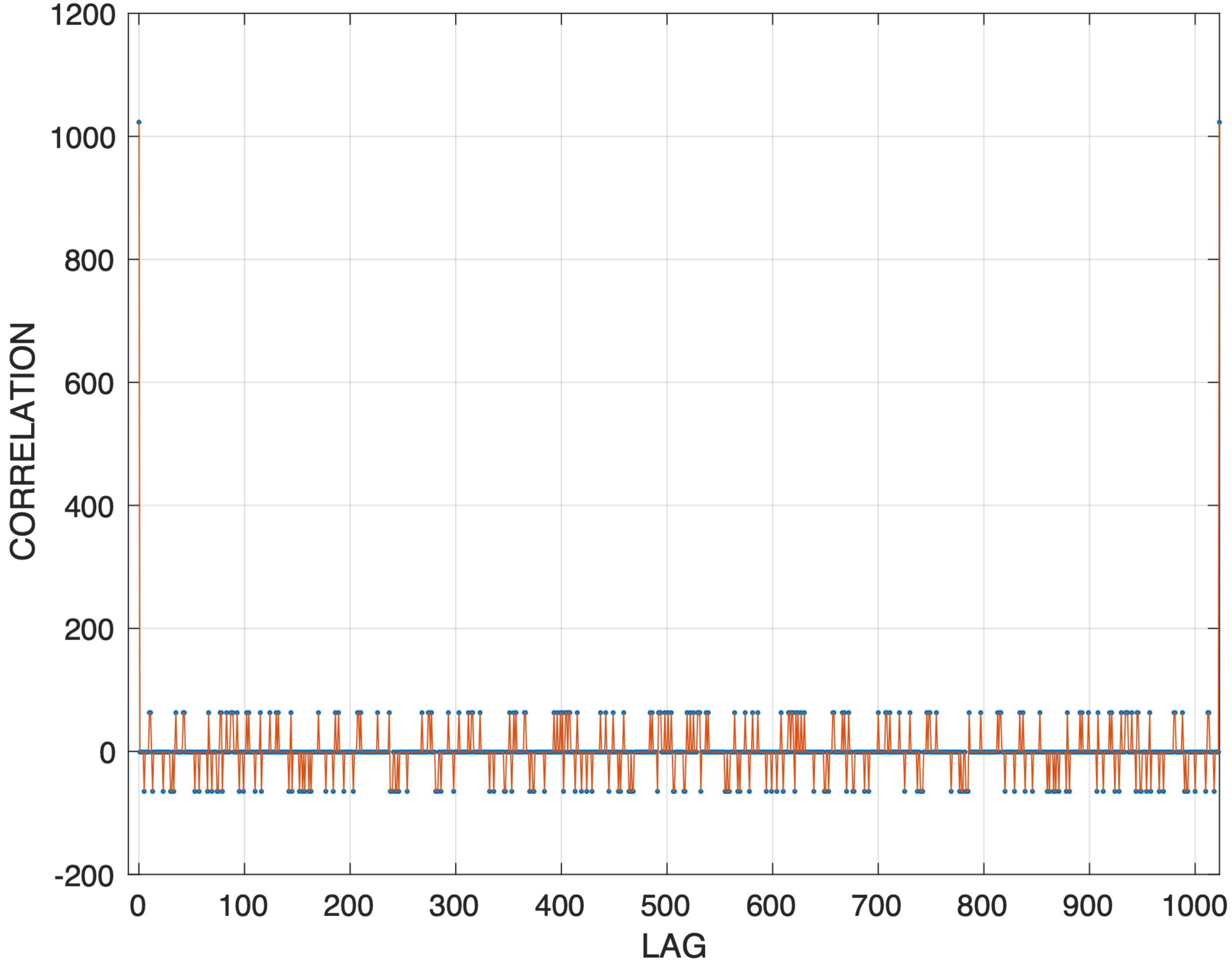
GPS C/A CODES (From Misra & Enge, Problem 2.1)



PRN 5 CODE SEQUENCE (shown as +/- 1's)



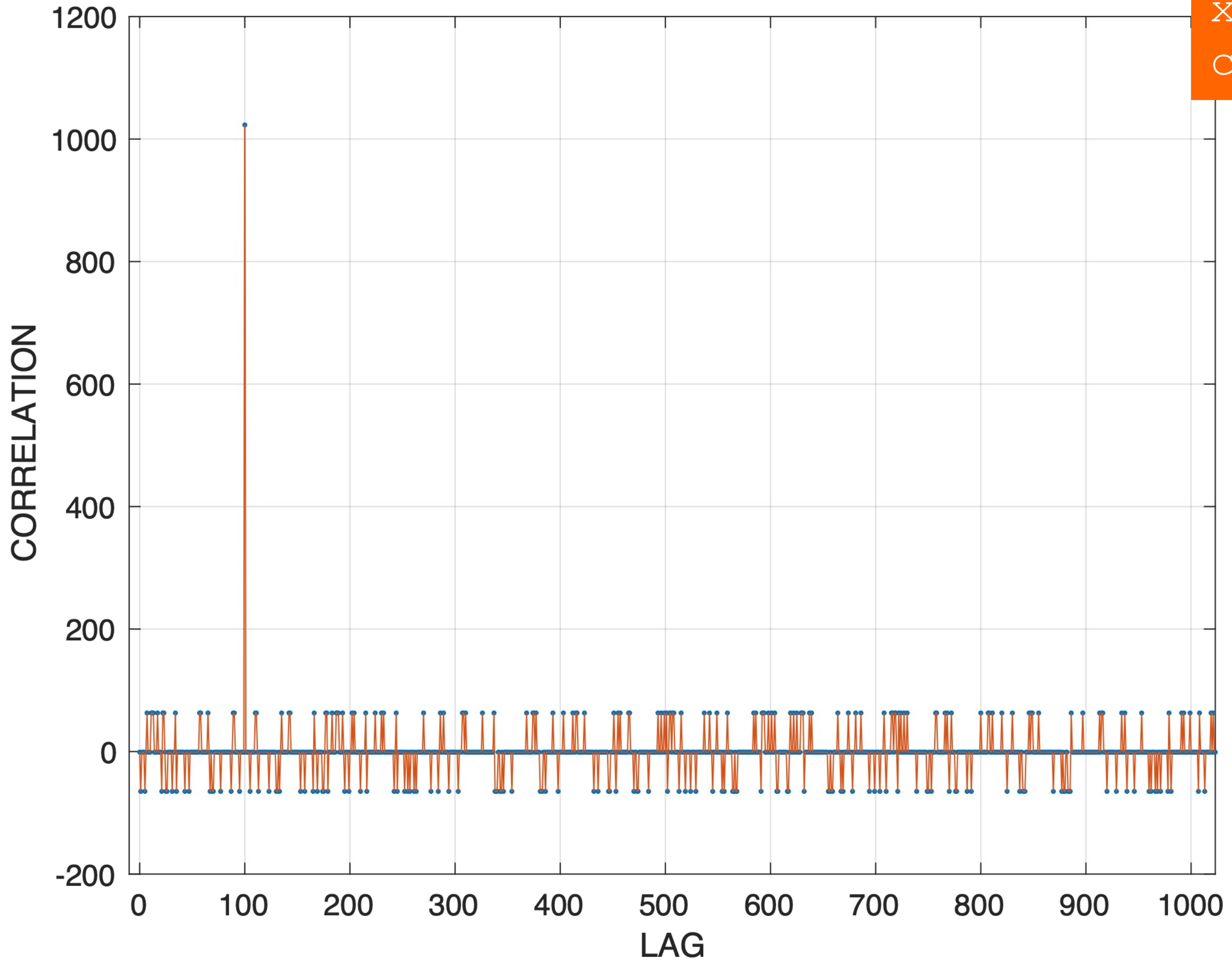
Autocorrelation of PRN 5 C/A Code



Computing a cyclic correlation

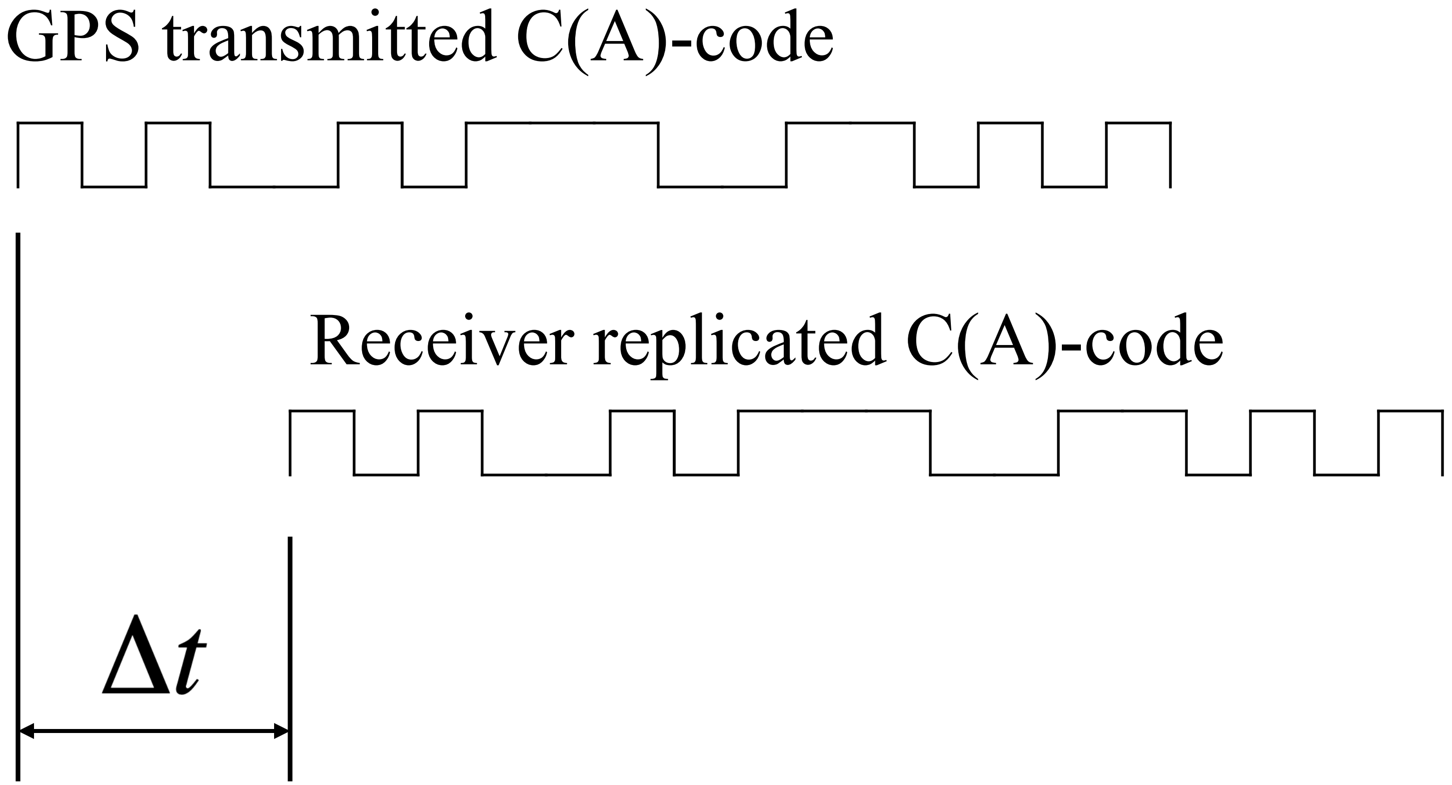
```
function P = cyc_corr_basic(x,y)
% Function to compute the cyclic correlation for ASEN 5090
% First input is the "received" signal
% Second input is the "replica" signal - this is the one that shifts

n = length(x);
x = reshape(x,1,n);          % make sure x is a row vector
yshifted = reshape(y,n,1);   % make the replica of y a column vector

lag = [0:n]; Rxy = NaN(size(lag));

for i = 1:length(lag)  % sweep the lag from 0 to n
    Rxy(i) = x*yshifted;
    yshifted = [yshifted(end); yshifted(1:end-1)];
end
plot(lag,Rxy,'.',lag,Rxy,'-'),grid,xlim([0 n])
xlabel('LAG'),ylabel('CORRELATION')
```

Cross correlation between shifted PRN5 & reference PRN 5




```
xshifted = x([end-99:end 1:end-100];
cyc_corr_basic(xshifted,x)
```

Pseudorange



- Δt is proportional to the GPS-to-receiver range
- Four pseudorange measurements can be used to solve for receiver position

GPS Satellite CA Code Phase Assignments

PRN #	Phase selection	Code delay	1st 10 chips (Octal)
1	$2 \oplus 6$	5	1440
2	$3 \oplus 7$	6	1620
3	$4 \oplus 8$	7	1710
4	$5 \oplus 9$	8	1744
5	$1 \oplus 9$	17	1133
6	$2 \oplus 10$	18	1455
7	$1 \oplus 8$	139	1131
8	$2 \oplus 9$	140	1454
9	$3 \oplus 10$	141	1626
10	$2 \oplus 3$	251	1504
11	$3 \oplus 4$	252	1642

PRN #	Phase selection	Code delay	1st 10 chips (Octal)
12	$5 \oplus 6$	254	1750
13	$6 \oplus 7$	255	1764
14	$7 \oplus 8$	256	1772
15	$8 \oplus 9$	257	1775
16	$9 \oplus 10$	258	1776
17	$1 \oplus 4$	469	1156
18	$2 \oplus 5$	470	1467
19	$3 \oplus 6$	471	1633
20	$4 \oplus 7$	472	1715
21	$5 \oplus 8$	473	1746
22	$6 \oplus 9$	474	1763

Code Phase Assignments Continued

PRN #	Phase selection	Code delay	1st 10 chips (Octal)
23	1 ⊕ 3	509	1063
24	4 ⊕ 6	512	1706
25	5 ⊕ 7	513	1743
26	6 ⊕ 8	514	1761
27	7 ⊕ 9	515	1770
28	8 ⊕ 10	516	1774
29	1 ⊕ 6	859	1127
30	2 ⊕ 7	860	1453
31	3 ⊕ 8	861	1625
32	4 ⊕ 9	862	1712

PRN #	Phase selection	Code delay	1st 10 chips (Octal)
33	5 ⊕ 10	863	1745
34	4 ⊕ 10	950	1713
35	1 ⊕ 7	947	1134
36	2 ⊕ 8	948	1456
37	4 ⊕ 10	950	1713

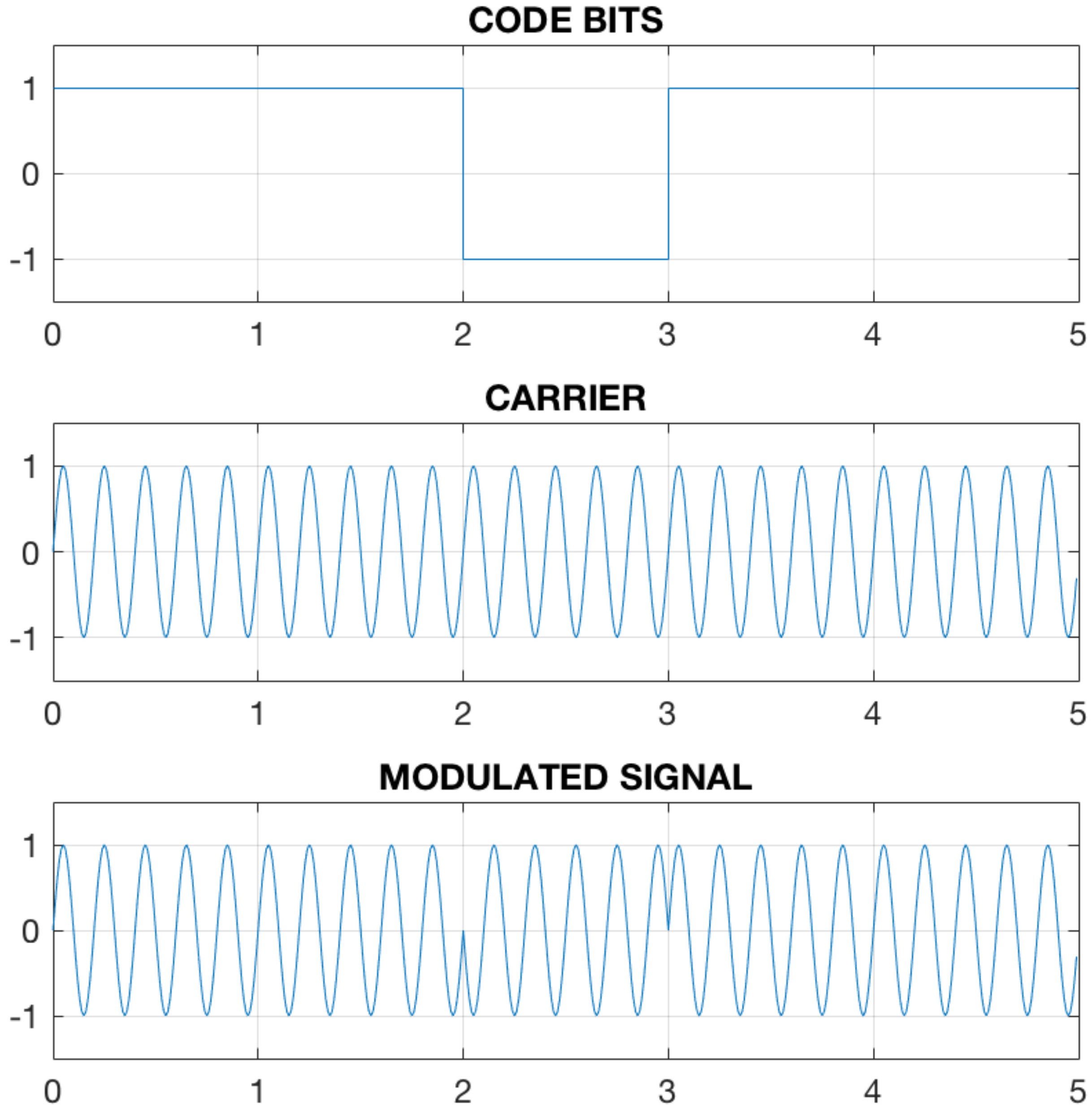
PRN# 33-37 are reserved for ground transmitters to validate system operation and accuracy

Impact of Correlation Properties

- What makes a useful code?
  - For ranging:
  - For CDMA:
  - For acquisition:

How are GPS PRN codes sent on a carrier signal?

GPS uses BPSK - (Bi-phase shift keying)
Sequence of "1's" and 0's" that specify when phase of carrier should be reversed (1) or left alone (0).



Can consider as a sequence of **0's and 1’s** OR **+1 and -1's**

- In terms of phase-shift keying:
  0 = "don't shift"
  1 = "shift by 180 deg"
- In terms of amplitude modulation:
  multiplying by "1" keeps the carrier
  multiplying by a "-1" inverts it

Binary Phase Shift Keying (BPSK) Modulation

Radio wave carrier: $s_c(t) = A\cos(\omega_c t + \varphi)$ (1)

BPSK modulated version is: $s_{cm}(t) = A\cos(\omega_c t + \varphi + \theta(t))$ (2)

Bi-phase modulation: $\theta(t) = 0$ or π

$$s_{cm}(t) = \begin{cases} A\cos(\omega_c t + \varphi) & \theta(t) = 0 \\ -A\cos(\omega_c t + \varphi) & \theta(t) = \pi \end{cases} \quad (3)$$

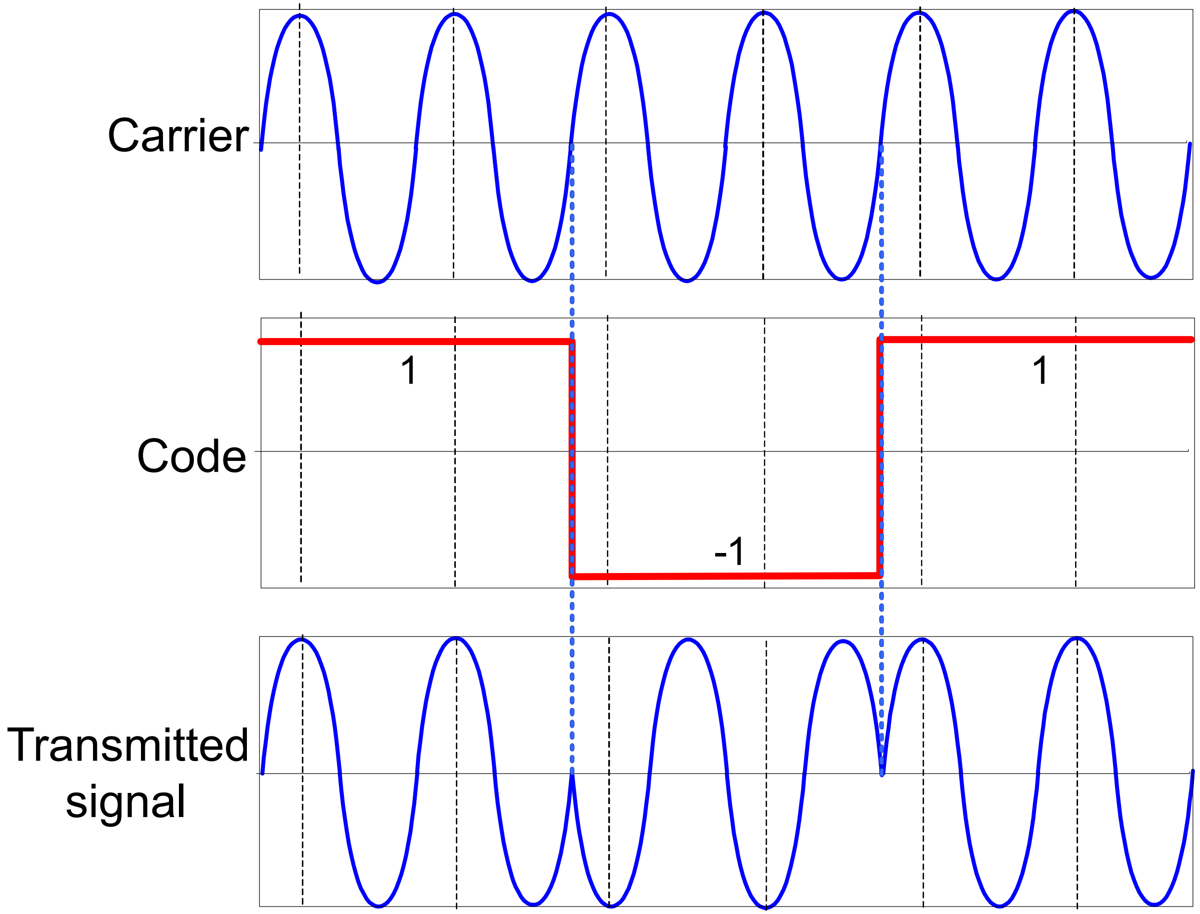
BPSK modulation ←→Binary Amplitude Shift Keying:

$$s_{cm}(t) = p(t)A\cos(\omega_c t + \varphi) \quad (4)$$

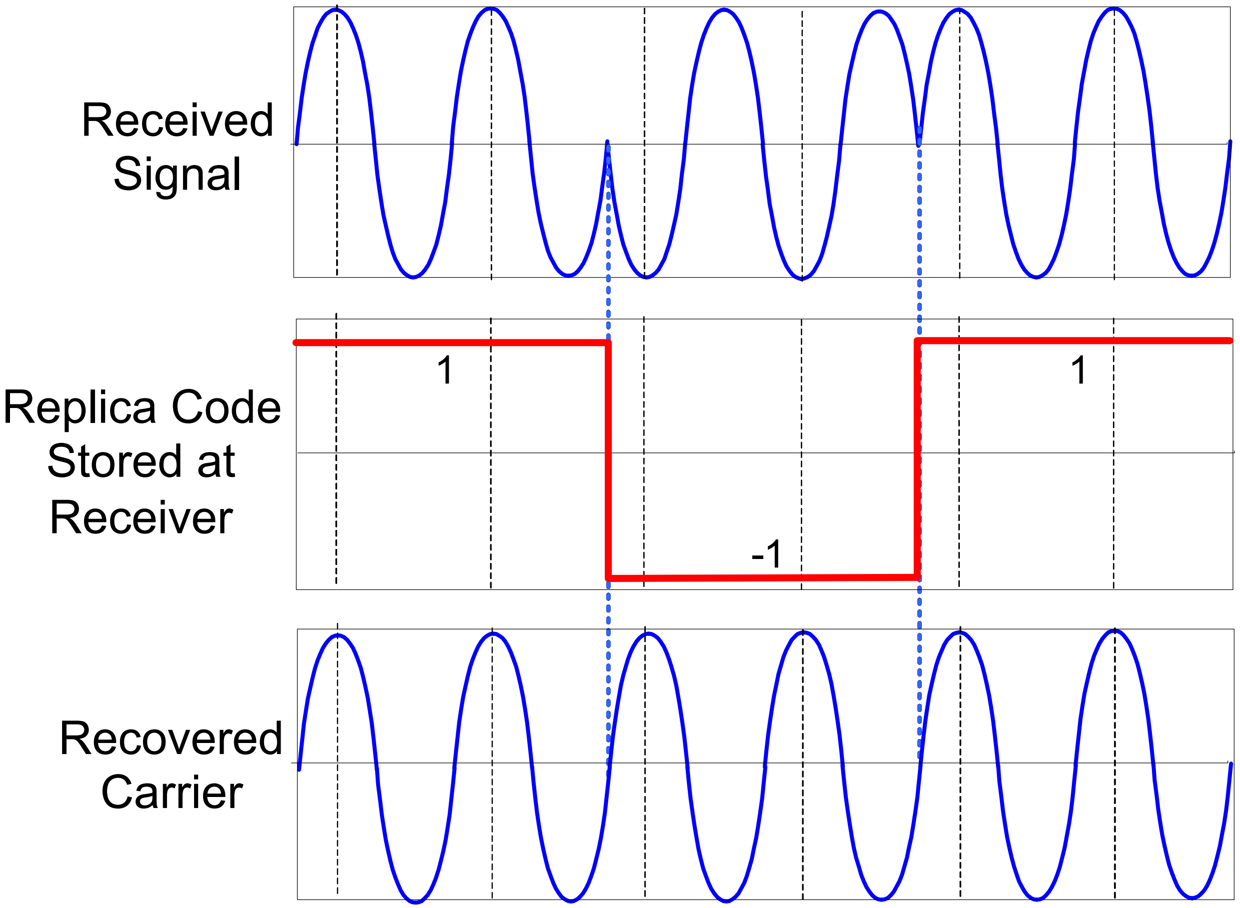
$p(t)$ = 1 or -1

BPSK Signal Transmission and Recovery

Transmitted from SV



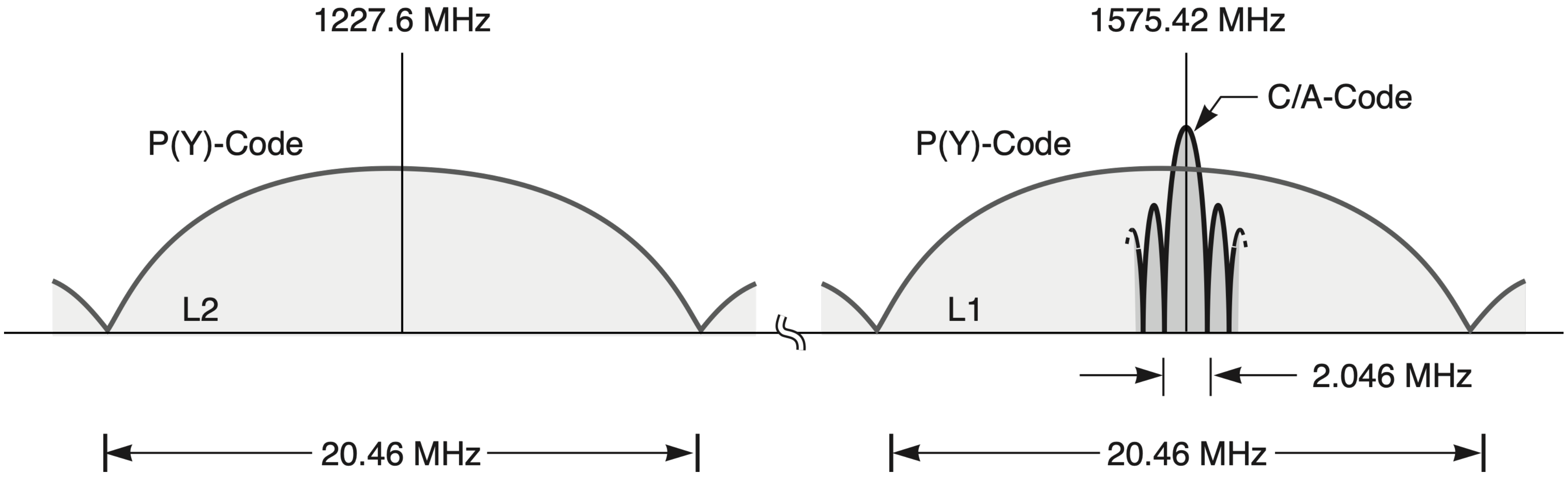
Recovered by Receiver



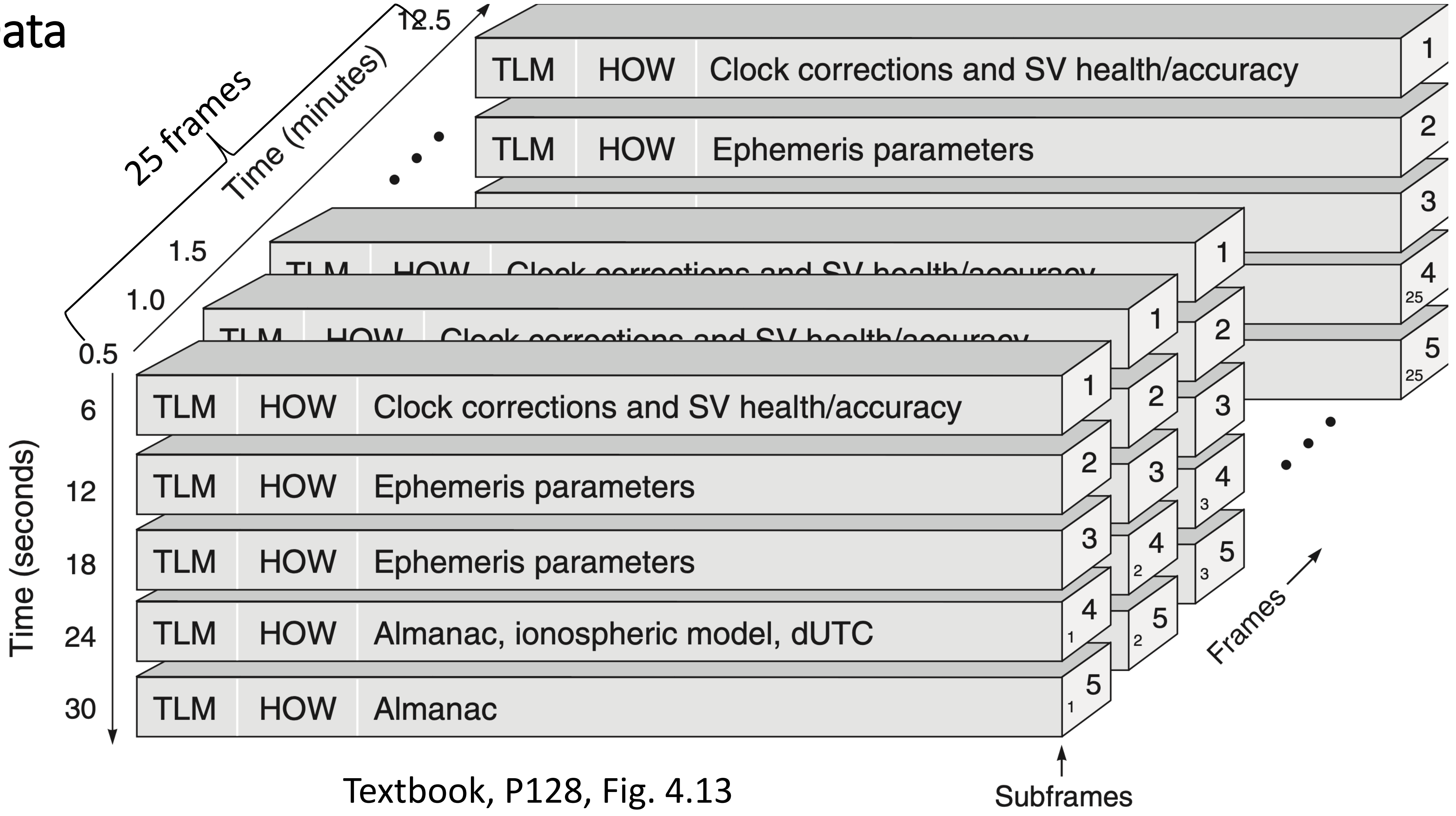
Effect of PRN Code Modulation on Spectrum

- LEGACY GPS

- PRN codes have a sinc^2 spectrum
- Main lobe is 2x chipping rate
- Side lobes are 1x chipping rate
- Power density is very low



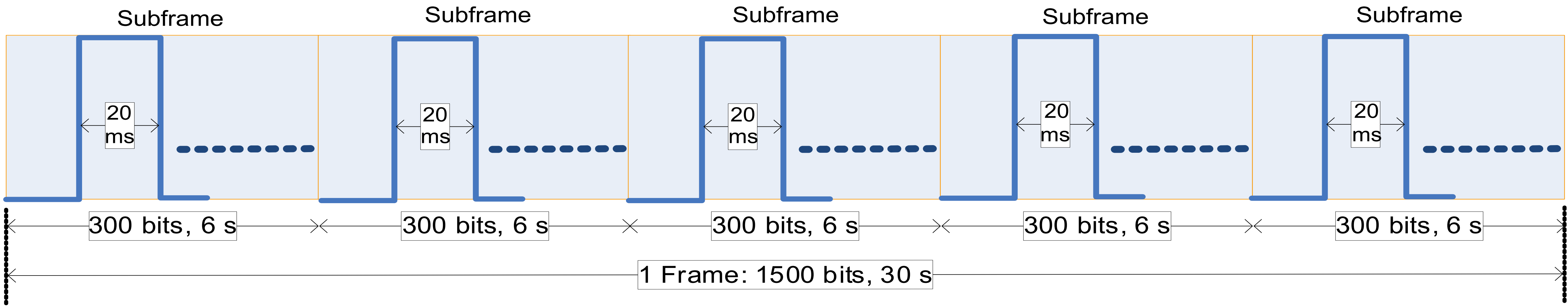
GPS Navigation Data Organization



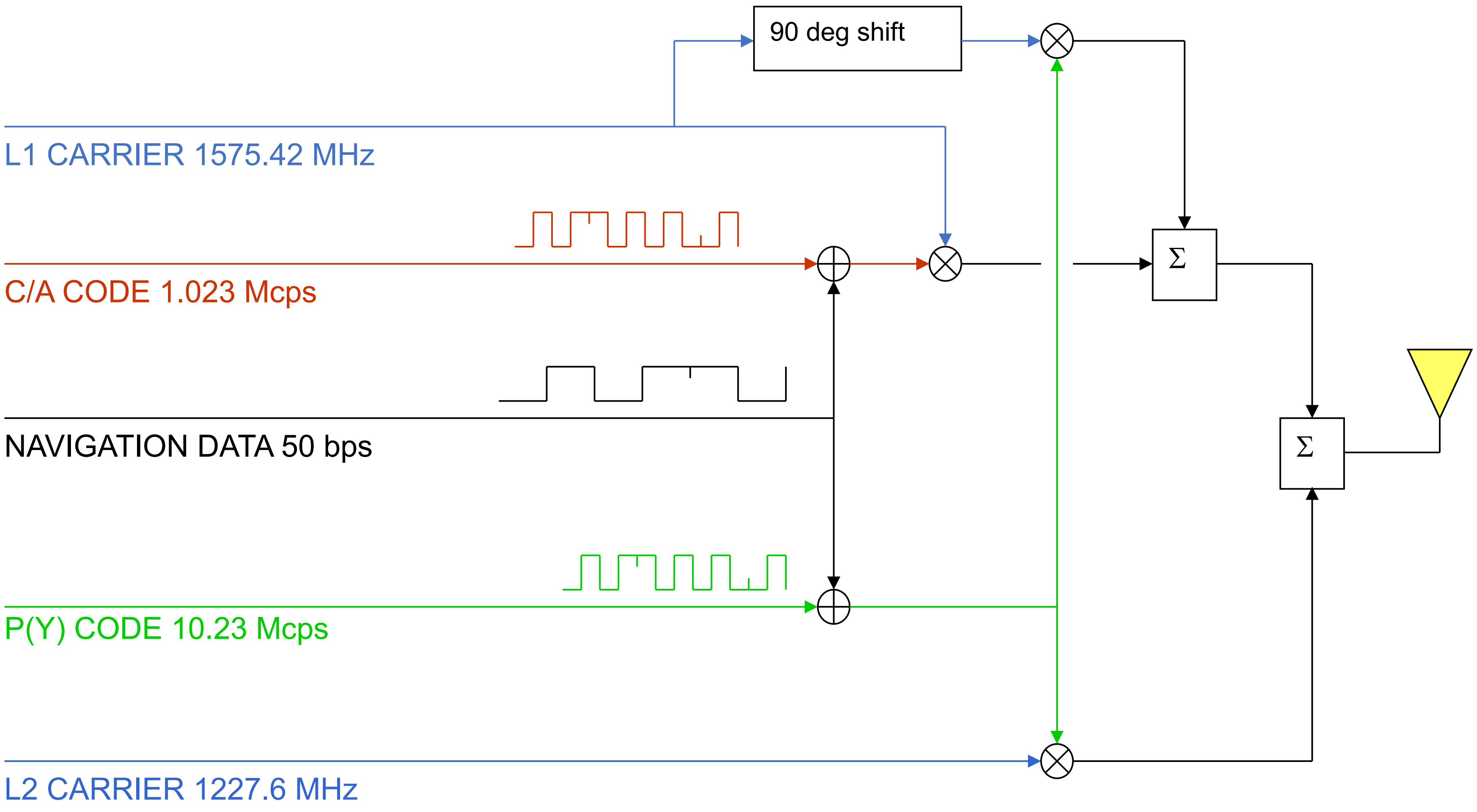
Textbook, P128, Fig. 4.13

TLM: telemetry word (synch pattern) HOW: hand-over word (contains Z-count)

Navigation Message: Time Sequence View within 1 Frame

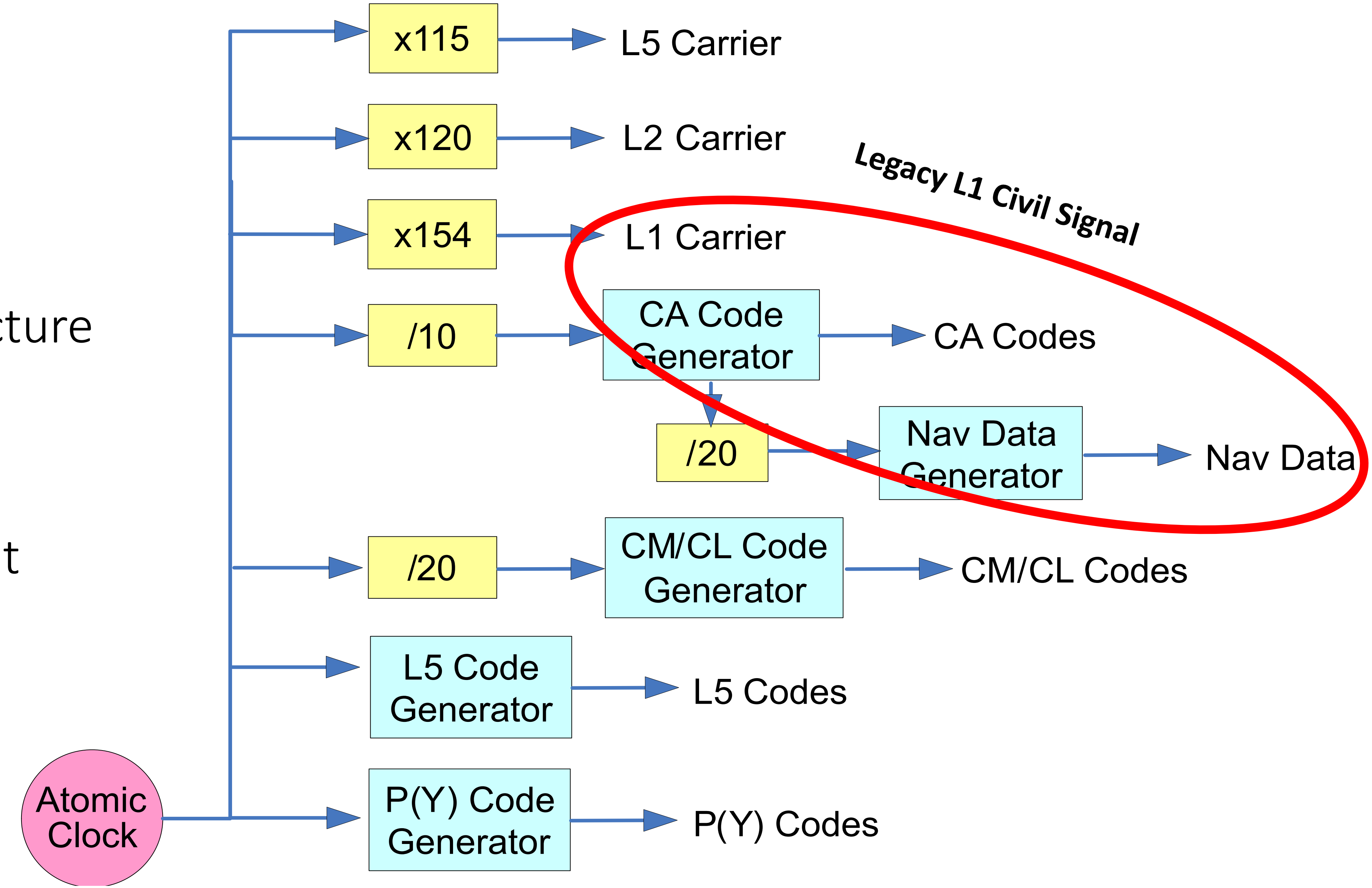


GPS LEGACY SIGNAL GENERATION



GPS Signal Structure Overview:

Frequencies of each component



Acknowledgements

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NEXT TIME – Coordinate Frames